PROGRAM 1: Streamlit Application for Text Generation

AIM

To develop a Streamlit application for text generation using a pretrained language model.

ALGORITHM

1. Import Streamlit and Transformers.
2. Load a pretrained text-generation model.
3. Accept a prompt from the user.
4. Generate text using the model.
5. Display the generated text.

CODE

```
import streamlit as st
from transformers import pipeline

st.title("Text Generation Application")

generator = pipeline(
    "text-generation",
    model="distilgpt2"
)

prompt = st.text_input("Enter your prompt:")

if st.button("Generate"):

    if prompt:
        result = generator(
            prompt,
            max_new_tokens=50,
            do_sample=True,
            pad_token_id=50256
```

```
    )


    st.write(result[0]["generated_text"])
else:
    st.warning("Please enter a prompt.")
```

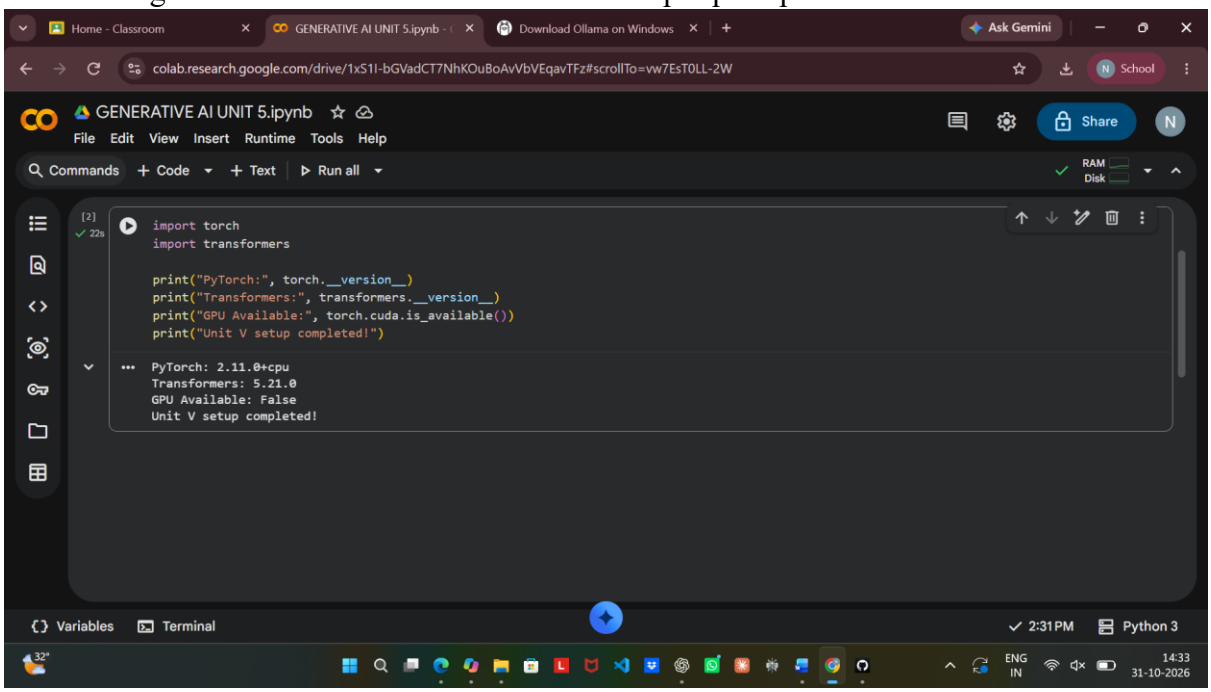
Execution note: This creates the Streamlit application code. To launch it from Colab, you need a Streamlit tunnel or a local runtime. For the simplest lab demonstration, you can test the same model directly in the next cell.

INPUT

Artificial Intelligence is

OUTPUT

The model generates additional text based on the input prompt.



RESULT

Thus, a text-generation application was developed using a pretrained language model.